

12/8/26

Activity - 05

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given input {4371, 1323, 6173, 4199, 4344, 9679, 1989}
and a hash function $f(x) = x \pmod{10}$, Show the
resulting

- separate chaining hash table
- open addressing hash table using linear probing
- open addressing hash table using quadratic probing
- Open addressing hash table with second hash

function $h_2(x) = 7 - (x \pmod{7})$

a) Separate chaining:-

$$h(x) = x \pmod{10}$$

$$h(4371) = 4371 \pmod{10} = 1$$

$$h(1323) = 1323 \pmod{10} = 3$$

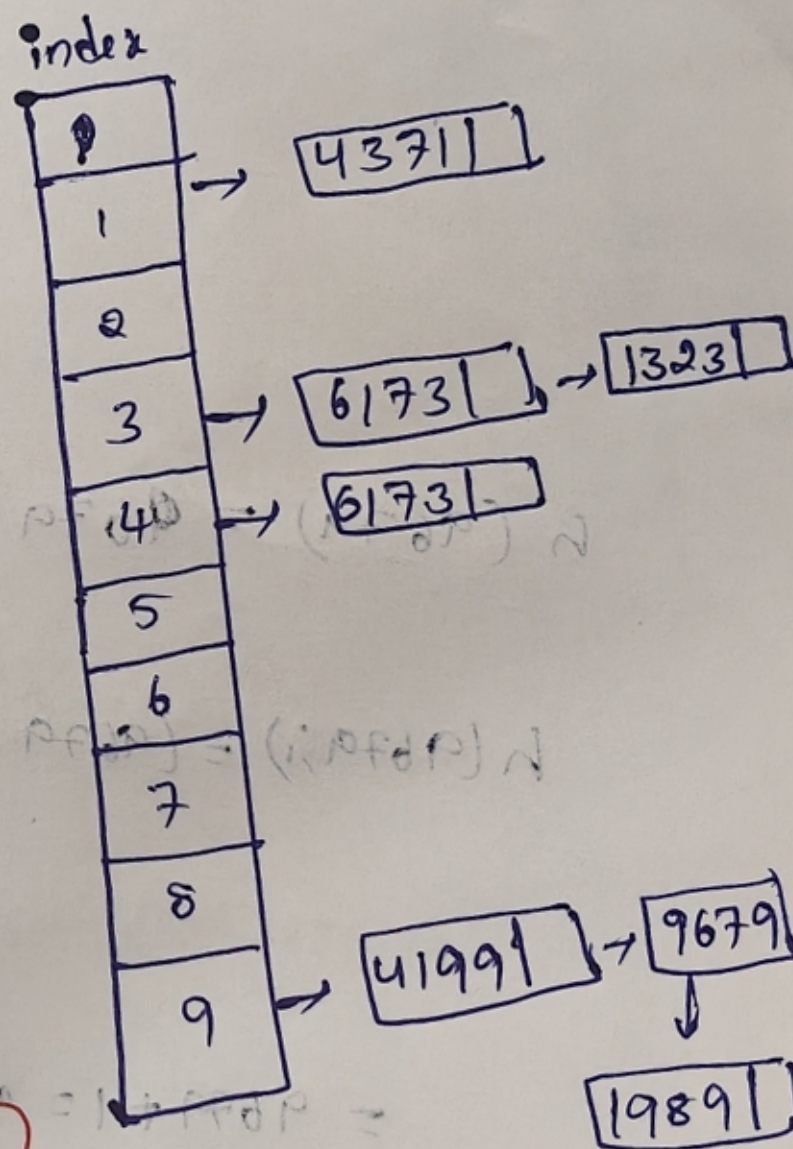
$$h(6173) = 6173 \pmod{10} = 3$$

$$h(4199) = 4199 \pmod{10} = 9$$

$$h(4344) = 4344 \pmod{10} = 4$$

$$h(9679) = 9679 \pmod{10} = 9$$

$$h(1989) = 1989 \pmod{10} = 9$$



b) Linear hashing:-

$$h(x) = h(\text{key} + i)$$

$$h(4371) = 4371 \pmod{10} = 1$$

$$h(1323) = 1323 \pmod{10} = 3$$

$$h(6173) = 6173 \pmod{10} = 3$$

$$h(6173) = 6173 \times 10^{-3} \text{ collision}$$

$$h(6173, i) = (6173 + 0) = 6173 \times 10^{-3}$$

$$\Rightarrow p = 1 \text{ collision}$$

$$= (6173 + 1) = 6174 \times 10^{-3}$$

$$\Rightarrow p = 0$$

$$h(4199) = 4199 \times 10^{-3}$$

$$h(4344) = 4344 \times 10^{-3} \text{ collision}$$

$$h(4344, i) = (4344 + 0) = 4344 \times 10^{-3}$$

$$\Rightarrow p = 1 \text{ collision}$$

$$= (4344 + 1) = 4345 \times 10^{-3}$$

$$\Rightarrow p = 2$$

$$h(9679) = 9679 \times 10^{-3} \text{ collision}$$

$$h(9679, i) = (9679 + 0) = 9679 \times 10^{-3}$$

$$\Rightarrow p = 1 \text{ collision}$$

$$= 9679 + 1 = 9680 \times 10^{-3}$$

$$\Rightarrow p = 2$$

$$h(1989) = 1989 \times 10^{-3} \text{ collision}$$

$$h(1989, i) = (1989 + 0) = 1989 \times 10^{-3}$$

$$\Rightarrow p = 1 \text{ collision}$$

$$= (1989 + 1) = 1990 \times 10^{-3}$$

$$\Rightarrow p = 0 \text{ collision}$$

0	→ 9679
1	→ 4344
2	→ 1989
3	→ 1321
4	→ 6173
5	→ 4344
6	
7	
8	
9	→ 4199

$$(1982 + 2) \Rightarrow 1991 \times 10 = 1$$

$\Rightarrow p=3$ collision

$$2 (1989 + 3) \Rightarrow 1992 \times 10 = 2$$

$\Rightarrow p=4$

c) Quadratic hashing

$$h(\text{key}) = h(\text{key} + i^2)$$

$\hookrightarrow i$ to table size

$$h(4371) = 4371 \times 10 = 1$$

$$h(1323) = 1323 \times 10 = 3$$

$$h(6173) = 6173 \times 10 = 3$$

collision

$$h(6173) = (6173 + 1) = 6174 \times 10 = 4$$

$\Rightarrow p=1$

$$h(4199) = 4199 \times 10 = 9$$

$$h(4344) = 4344 \times 10 = 4 \text{ collision}$$

$$h(4344, i) = (4344 + 1) = 4345 \times 10 = 5$$

$\Rightarrow p=1$

$$h(9679) = 9679 \times 10 = 9$$

collision

$$h(9679, i) = (9679 + 1) = 9680 \times 10 = 0$$

$\Rightarrow p=1$

$$h(1989) = 1989 \times 10 = 9 \text{ collision}$$

$$h(9679, i) = (1989 + 1) = 1990 \times 10 = 0$$

$\Rightarrow p=1$ collision

$$h(1989 + 4) = 1993 \times 10 = 3$$

$\Rightarrow p=2$ collision

$$h(1989 + 9) = 1998 \times 10 = 8$$

$\Rightarrow p=3$

0	→ 9679
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 4344
6	
7	
8	→ 1989
9	→ 4199

D) Double hashing:

$$h(\text{key}) = [h_1(\text{key}) + (h_2(\text{key}) * i)] \therefore R = \text{prime number}$$

$$h_1(\text{key}) = x \pmod{t_3} \quad h_2(\text{key}) = R - x \pmod{R}$$

$$R = 10 \Rightarrow 2, 3, 5, 7$$

$$R = 7$$

$$h(4371) = 4371 \times 10 = 43710$$

$$h(1323) = 1323 \times 10 = 13230$$

$$h(6173) = 6173 \times 10 = 61730$$

$$h(6173) = 6173 \times 10 + (7 - 6173 \times 10) * 1$$

$$= 3 + (4) = 7$$

$$h(4199) = 4199 \times 10 = 41990$$

$$h(4344) = 4344 \times 10 = 43440$$

$$h(9679) = 9679 \times 10 = 96790 \text{ collision}$$

$$h(9679) = 9679 \times 10 + (7 - 9679 \times 10) * 1$$

$$= 9 + (-2) = 7$$

$$= 9679 \times 10 + (7 - 9679 \times 10) * 2$$

$$= 9 + (-2)(2) = 5$$

$$h(1989) = 1989 \times 10 = 19890 \text{ collision}$$

$$= 1989 \times 10 + (7 - 1989 \times 10) * 1$$

$$= 9 + (7 - 1) * 1 = 9$$

$$= 9 + 6 = 15 \times 10 = 5 \text{ collision}$$

0	
1	→ 4371
2	
3	→ 1323
4	→ 6173
5	→ 9679
6	
7	→ 4344
8	
9	→ 4199

$$= 9 + (7-1) * 2$$

$$= 9 + 12 = 21 \% 10 = 1 \text{ collision}$$

$$= 9 + (7-1) * 3$$

$$= 9 + 18 = 27 \% 10 = 7 \text{ collision}$$

$$= 9 + (7-1) * 4$$

$$= 9 + 24 = 33 \% 10 = 3 \text{ collision}$$

$$= 9 + (7-1) * 5$$

$$= 9 + 30 = 39 \% 10 = 9 \text{ collision}$$

$$= 9 + (7-1) * 6$$

$$= 9 + 36 = 45 \% 10 = 5 \text{ collision}$$

$$= 9 + (7-1) * 7$$

$$= 51 \% 10 = 1 \text{ collision}$$

$$= 9 + (7-1) * 8$$

$$= 57 \% 10 = 7 \text{ collision}$$

Note:

1989
position

hash value

because

doesn't have index

value has extend the

fabular value